Inspiring Excellence

CSE 421 Lab Project
Section: 01
Semester: Fall 2025

Submitted by : Group 910

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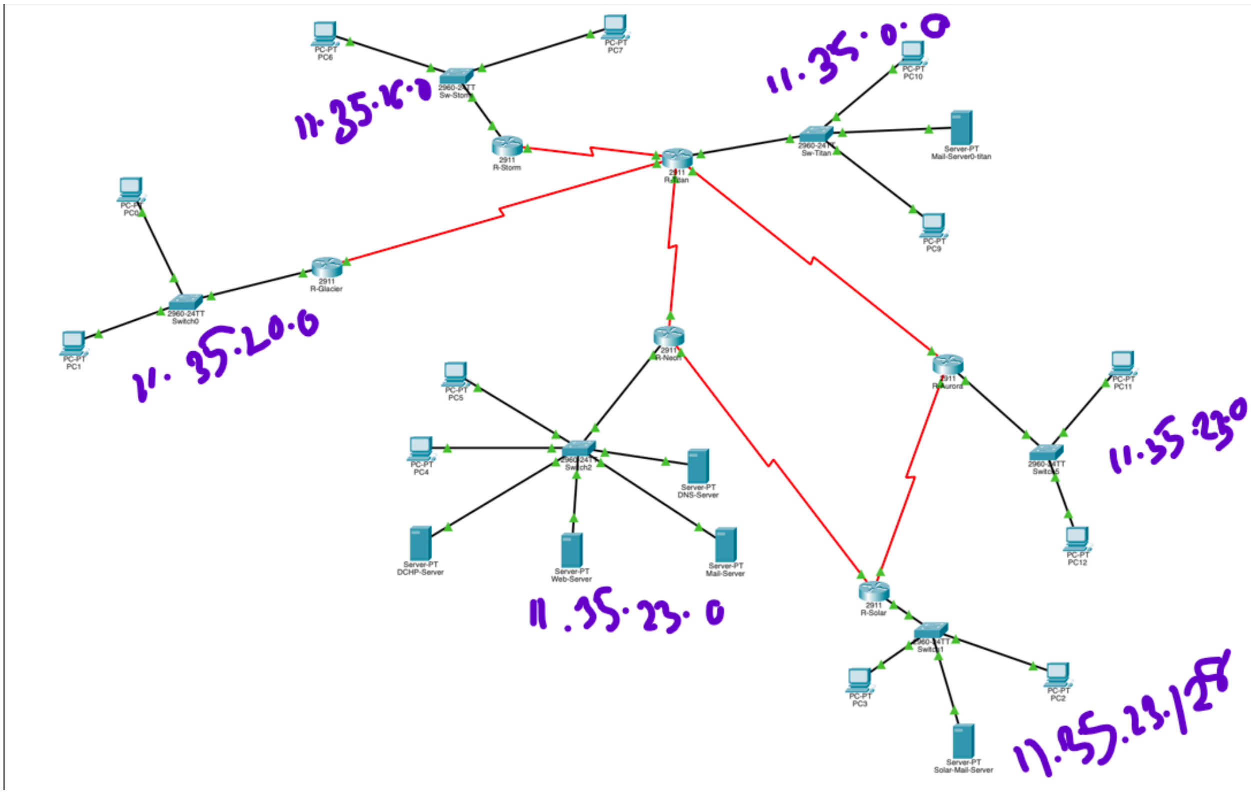
Topology:



VLSM Tree / Subnet Calculation

Root Network: 11.35.0.0/16 (Used as the base address for the Crystal Empire Network)

A. WAN Link Subnets (Infrastructure)

- Block 1: 11.35.23.192/30—Titan Valley ↔ Neon Hills

    - Block 2: 11.35.23.196/30—Titan Valley ↔ Glacier Fields
    - Block 3: 11.35.23.200/30—Titan Valley ↔ Storm Harbor
    - Block 4: 11.35.23.204/30—Titan Valley ↔ Aurora Ridge
    - Block 5: 11.35.23.208/30—Aurora Ridge ↔ Solar Palace
    - Block 6: 11.35.23.212/30 —- Neon Hills — Solar Palace

- Block 6

B. LAN Subnets (Stations)

- **11.35.0.0 /20—Titan Valley LAN**
- **11.35.16.0 /22—Storm Harbor LAN**
- **11.35.20.0/23—Glacier Fields LAN**
- **11.35.22.0/24—Aurora Ridge LAN**
- **11.35.23.0 /25—Neon Hills LAN (Servers)**
- **11.35.23.128/26—Solar Palace LAN**

Detailed IP Addressing Table

Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Role / Notes
R-TITAN VALLEY	G0/0	11.35.0.1	255.255.240.0	N/A	LAN Gateway (Titan Valley)
	S0/0/0	11.35.23.201	255.255.255.252	N/A	Link to Storm Harbor
	S0/0/1	11.35.23.197	255.255.255.252	N/A	Link to Glacier Fields
	S0/1/0	11.35.23.193	255.255.255.252	N/A	Link to Neon Hills
	S0/1/1	11.35.23.205	255.255.255.252	N/A	Link to Aurora Ridge

Titan Valley PC9	NIC	11.35.0.2	255.255.240.0	11.35.0.1	Static Station PC
Titan Valley PC10	NIC	11.35.0.3	255.255.240.0	11.35.0.1	Static Station PC
TITAN VALLEY Mail Server	NIC	11.35.0.4	255.255.240.0	11.35.0.1	Handles Emails
R-STORM HARBOR	G0/0	11.35.16.1	255.255.252.0	N/A	LAN Gateway (Storm Harbor)
	S0/0/0	11.35.23.202	255.255.255.252	N/A	Link to Titan Valley
STORM HARBOR PC6	NIC	11.35.16.2	255.255.252.0	11.35.16.1	Static PC
STORM HARBOR PC7	NIC	11.35.16.3	255.255.252.0	11.35.16.1	Static PC
R-GLACIER FIELDS	G0/0	11.35.20.1	255.255.254.0	N/A	LAN Gateway (Glacier Fields)
	S0/0/0	11.35.23.198	255.255.255.252	N/A	Link to Titan Valley

GLACIER FIELDS PC0	NIC	11.35.20.2	255.255.254.0	11.35.20.1	Static PC
GLACIER FIELDS PC1	NIC	11.35.20.3	255.255.254.0	11.35.20.1	Static PC
R-NEON HILLS	G0/0	11.35.23.1	255.255.255.128	N/A	LAN Gateway (Neon Hills)
	S0/0/0	11.35.23.194	255.255.255.252	N/A	Link to Titan Valley
NEON HILLS PC4	NIC	11.35.23.2	255.255.255.128	11.35.23.1	Static PC
NEON HILLS PC5	NIC	11.35.23.3	255.255.255.128	11.35.23.1	Static PC
NEON HILLS DHCP Server	NIC	11.35.23.4	255.255.255.128	11.35.23.1	Provides IPs
NEON HILLS WEB Server	NIC	11.35.23.5	255.255.255.128	11.35.23.1	Hosts Neon Website
NEON HILLS MAIL Server	NIC	11.35.23.6	255.255.255.128	11.35.23.1	Handles Emails

NEON HILLS DNS Server	NIC	11.35.23.7	255.255.255.128	11.35.23.1	Resolves Names

R-AURORA RIDGE	G0/0	11.35.22.1	255.255.255.0	N/A	LAN Gateway (Aurora Ridge)
	S0/0/0	11.35.23.206	255.255.255.252	N/A	Link to Titan Valley
	S0/0/1	11.35.23.209	255.255.255.252	N/A	Link to Solar Palace
AURORA RIDGE PC11	NIC	11.35.22.2	255.255.255.0	11.35.22.1	Static Station PC
AURORA RIDGE PC12	NIC	11.35.22.3	255.255.255.0	11.35.22.1	Static Station PC
R-SOLAR PALACE	G0/0	11.35.23.129	255.255.255.192	N/A	LAN Gateway (Solar Palace)
	S0/0/0	11.35.23.210	255.255.255.252	N/A	Link to Aurora Ridge

SOLAR PALACE PC2	NIC	11.35.23.130	255.255.255.192	11.35.23.129	Static PC
SOLAR PALACE Mail Server	NIC	11.35.23.132	255.255.255.192	11.35.23.129	Handles Emails
SOLAR PALACE PC3	NIC	11.35.23.131	255.255.255.192	11.35.23.129	Static PC

6. List of Assumptions

1. Subnet Size Selection Using VLSM:

The base network 11.35.0.0/16 was subnetted using VLSM based on the population size of each location. Larger regions such as Titan Valley and Storm Harbor were allocated larger subnets, while smaller regions such as Solar Palace and Neon Hills were assigned smaller subnets to minimize IP address wastage while still meeting host requirements.

2. Neon Hills Server Addressing Assumption:

All core services (Web, DNS, Email, and DHCP) were assumed to be hosted in Neon Hills and assigned static IP addresses. This ensures consistent accessibility, reliable name resolution, and stable service operation across the entire Crystal Empire network.

3. DHCP Scope and Relay Configuration:

It was assumed that Glacier Fields and Aurora Ridge would not use static IP addressing for end devices. Instead, their IP addresses are dynamically assigned from the DHCP server located in Neon Hills, with DHCP relay (ip helper-address) configured on their respective routers to forward DHCP requests.

4. Floating Static Route Behavior:

A floating static route was configured with a higher administrative distance than RIP v2 to act strictly as a backup path. It is assumed that this route remains inactive during normal operation and only becomes active if the primary dynamically learned route fails, ensuring redundancy without causing routing loops.

5. Static and Dynamic Routing Coexistence:

The network was assumed to operate as a hybrid routing environment, where specific paths use static routing as required (Solar Palace ↔ Neon Hills and Neon Hills ↔ Titan Valley), while all other routes are learned dynamically.

6. Recursive Static Routing Path Assumption:

For the Solar Palace ↔ Neon Hills communication, recursive static routing was assumed to function correctly by resolving the next-hop IP address through routes learned via RIP v2. This ensures reachability without specifying a directly attached exit interface.

7. Exit-Interface Static Route Stability:

The static routing between Neon Hills and Titan Valley was assumed to be stable due to their direct physical connection. Therefore, static routes using the directly attached exit interface were considered sufficient and reliable for this link.

8. Email Access Control Assumption:

Email communication was assumed to be restricted strictly to Solar Palace, Neon Hills, and Titan Valley. Access control lists (ACLs) were applied to enforce this policy, while other locations were assumed to have no email service access despite having full IP connectivity.

9. End Device Configuration Mode:

End devices in Solar Palace, Neon Hills, Titan Valley, and Storm Harbor were assumed to use manually assigned static IP addresses, while only Glacier Fields and Aurora Ridge relied on DHCP-based configuration.

Commands
Static:
----

neon-solar

R-Neon>en
R-Neon#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-Neon(config)#ip route 11.35.23.128 255.255.255.192 11.35.23.193 1
R-Neon(config)#
R-Neon(config)#
R-Neon(config)#interface Serial0/0/1
R-Neon(config-if)#ip address 11.35.23.213 255.255.255.252
R-Neon(config-if)#
R-Neon(config-if)#exit
R-Neon(config)#interface Serial0/0/1
R-Neon(config-if)#no shutdown
R-Neon(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
exit
R-Neon(config)#ip route 11.35.23.128 255.255.255.192 se0/0/1 5
R-Neon(config)#

R-Solar>enable
R-Solar#
R-Solar#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R-Solar(config)#interface Serial0/0/1
R-Solar(config-if)#ip address 11.35.23.214 255.255.255.252

R-Solar(config-if)#no shutdown
R-Solar(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

R-Solar(config-if)#exit
R-Solar(config)#ip route 11.35.23.0 255.255.255.128 11.35.23.193 1
R-Solar(config)#ip route 11.35.23.0 255.255.255.128 se0/0/1 5
R-Solar(config)#exit
R-Solar#

neon --- titan

R-Neon(config)#ip route 11.35.0.0 255.255.240.0 se0/0/0 1
R-Neon(config)#exit
R-Neon#

R-Titan>en
R-Titan#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-Titan(config)#ip route 11.35.23.0 255.255.255.128 se0/1/0 1
R-Titan(config)#exit
R-Titan#
%SYS-5-CONFIG_I: Configured from console by console

dynamic part:

R-Glacier : RIP config

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname R-Glacier
R-Glacier(config)#router rip
R-Glacier(config-router)#version 2
R-Glacier(config-router)#no auto-summary

```
R-Glacier(config-router)#network 11.35.20.0
R-Glacier(config-router)#network 11.35.0.0
R-Glacier(config-router)#passive-interface g0/0
R-Glacier(config-router)#default-information originate
R-Glacier(config-router)#end
R-Glacier#
%SYS-5-CONFIG_I: Configured from console by console
```

R-Storm : RIP config

```
R-Storm>en
R-Storm#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-Storm(config)#router rip
R-Storm(config-router)#version 2
R-Storm(config-router)#no auto-summary
R-Storm(config-router)#network 11.35.16.0
R-Storm(config-router)#network 11.35.0.0
R-Storm(config-router)#passive-interface g0/0
R-Storm(config-router)#default-information originate
R-Storm(config-router)#end
R-Storm#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R-Storm#
```

R-Aurora : RIP config

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R-Aurora
R-Aurora(config)#router rip
R-Aurora(config-router)#version 2
R-Aurora(config-router)#no auto-summary
R-Aurora(config-router)#network 11.35.22.0
R-Aurora(config-router)#network 11.35.0.0
R-Aurora(config-router)#network 11.35.23.128
R-Aurora(config-router)#passive-interface g0/0
R-Aurora(config-router)#default-information originate
R-Aurora(config-router)#end
R-Aurora#
```

%SYS-5-CONFIG_I: Configured from console by console

R-Solar : RIP config

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R-Solar
R-Solar(config)#router rip
R-Solar(config-router)#version 2
R-Solar(config-router)#no auto-summary
R-Solar(config-router)#network 11.35.23.128
R-Solar(config-router)#network 11.35.22.0
R-Solar(config-router)#passive-interface g0/0
R-Solar(config-router)#default-information originate
R-Solar(config-router)#end
R-Solar#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R-Solar#
R-Solar#
```

R-Titan : RIP config

```
R-Titan>en
R-Titan#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-Titan(config)#router rip
R-Titan(config-router)#version 2
R-Titan(config-router)#no auto-summary
R-Titan(config-router)#network 11.35.0.0
R-Titan(config-router)#network 11.35.20.0
R-Titan(config-router)#network 11.35.16.0
R-Titan(config-router)#network 1.35.23.0
R-Titan(config-router)#no network 1.35.23.0
R-Titan(config-router)#network 11.35.23.0
R-Titan(config-router)#network 11.35.22.0
R-Titan(config-router)#passive-interface g0/0
```

```
R-Titan(config-router)#default-information originate
R-Titan(config-router)#end
R-Titan#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R-Titan#
```

R-Titan : RIP config

```
R-Neon>en
R-Neon#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-Neon(config)#router rip
R-Neon(config-router)#no auto-summary
R-Neon(config-router)#network 11.35.0.0
R-Neon(config-router)#network 11.35.23.0
R-Neon(config-router)#passive-interface g0/0
R-Neon(config-router)#default-information originate
R-Neon(config-router)#end
R-Neon#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R-Neon#
```

DHCP Config Glacier and Aurora:

```
R-Neon>en
R-Neon#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-Neon(config)#ip dhcp excluded-address 11.35.20.1 11.35.20.10
R-Neon(config)#ip dhcp excluded-address 11.35.22.1 11.35.22.10
R-Neon(config)#ip dhcp pool R-Glacier-LAN
R-Neon(dhcp-config)#network 11.35.20.0 255.255.254.0
R-Neon(dhcp-config)#default-router 11.35.20.1
R-Neon(dhcp-config)#dns-server 11.35.23.7
R-Neon(dhcp-config)#exit
```

R-Neon(config)#ip dhcp pool R-Aurora-LAN
R-Neon(dhcp-config)#network 11.35.22.0 255.255.255.0
R-Neon(dhcp-config)#default-router 11.35.22.1
R-Neon(dhcp-config)#dns-server 11.35.23.7
R-Neon(dhcp-config)#exit
R-Neon(config)#

Relay agent

R- Glacier:

R-Glacier>en
R-Glacier#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-Glacier(config)#int g0/0/0
%Invalid interface type and number
R-Glacier(config)#int g0/0
R-Glacier(config-if)#ip helper-address 11.35.23.194
R-Glacier(config-if)#exit
R-Glacier(config)#

R-Aurora:

R-Aurora>en
R-Aurora#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-Aurora(config)#int g0/0
R-Aurora(config-if)#ip helper-address 11.35.23.194
R-Aurora(config-if)#exit
R-Aurora(config)#
R-Titan:

R-Titan>en
R-Titan#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-Titan(config)#int se0/0/
                        ^
% Invalid input detected at '^' marker.

R-Titan(config)#int se0/0/1
R-Titan(config-if)#ip helper-address 11.35.23.194

```
R-Titan(config-if)#exit
R-Titan(config)#int se0/1/1
R-Titan(config-if)#ip helper-address 11.35.23.194
R-Titan(config-if)#exit
R-Titan(config)#exit
R-Titan#
%SYS-5-CONFIG_I: Configured from console by console
```

WEB server

```
<html>
<head>
<title>The Crystal Empire Networ</title>
</head>
<body>
<h1>Welcome to the Crystal Empire of AURORIA</h1>
</body>
</html>
```

mail service:

```
R-Neon>en
R-Neon#conf t
Enter configuration commands, one per line.  End with CNTL/Z.

R-Neon(config)#access-list 110 permit tcp 11.35.23.128 0.0.0.63 host 11.35.23.6
eq 25
R-Neon(config)#access-list 110 permit tcp 11.35.0.0 0.0.15.255 host 11.35.23.6
eq 25
R-Neon(config)#access-list 110 permit tcp 11.35.23.0 0.0.0.127 host 11.35.23.6
eq 25
R-Neon(config)#access-list 110 permit tcp 11.35.23.128 0.0.0.63 host 11.35.23.6
eq 110
R-Neon(config)#access-list 110 permit tcp 11.35.0.0 0.0.15.255 host 11.35.23.6
eq 110
R-Neon(config)#access-list 110 permit tcp 11.35.23.0 0.0.0.127 host 11.35.23.6
eq 110
R-Neon(config)#access-list 110 permit ip any any
R-Neon(config)#interface GigabitEthernet0/0
R-Neon(config-if)# ip access-group 110 in
```

```
R-Neon(config-if)#ip access-group 110 in
R-Neon(config-if)#end
```

Work distribution:

NAME: NISHAT ZAHAN NIHA, ID: 22201048

1. Neon Hills Server Deployment

2. Email Server
3. Restricted Email Service Setup
4. Router Interface Configuration
5. Network Testing & Verification
6. Report

NAME: SADIA HABIB NIZHUM, ID: 24241224

1. Topology
2. VLSM Tree
3. Ip addressing
4. LAN & End Device Configuration
5. Router Interface Configuration
6. Network Testing & Verification
7. Report

NAME: SHAIKH SAMMO AL AZIZ, ID: 24141135

1. Static Routing Configuration, RIP v2 Dynamic Routing Setup

2. Floating Static Route Implementation
3. DHCP Server
4. Web & DNS Service Configuration

5. Network Testing & Verification

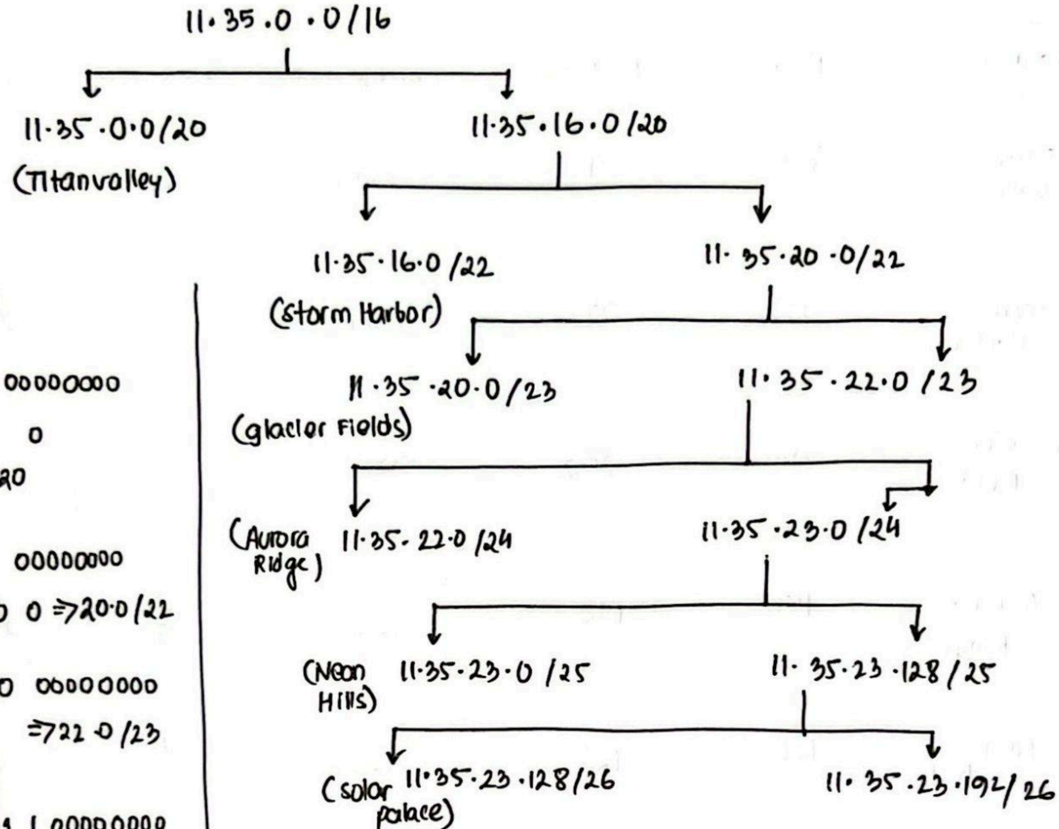
6. Router Interface Configuration
7. Report

The lost four digits of the last member is 1135

∴ The Base Network will be 11.35.0.0/16

Location	Host	Host +2	Block size	Host bits	Prefix
Titan valley	4050	4052	4096	12	20
storm Harbor,	900	902	1024	10	22
Glacier Fields	350	352	512	9	23
Aurora Ridge	150	152	256	8	24
Neon Hills	126	128	128	7	25
solar Palace	40	42	64	6	26

Subnetting tree



Rough

10000000 00000000

0001 0000 0

16·0/20

0001|00|00 00000000

0001|01|0|0 0 => 20·0/22

000101| 1 |0 00000000

=> 22·0/23

0001 011 |1| 00000000

=> 23·0/24

23 |1|0| 000000

=> 128

23 1 |1| 000000

=> 23·192